Discussion 6: Real-time Autonomous Systems

Discussion Topic:

As technology evolves, so do the challenges associated with real-time systems. Modern advancements in computing, such as autonomous systems, edge computing, and 5G networks, have introduced new real-time challenges that need innovative solutions.

For this week's discussion, you are tasked with identifying a current real-time system challenge and exploring its relevance to modern computing. Some potential areas include:

- The integration of AI in real-time decision-making systems
- Edge computing and how real-time constraints affect distributed systems
- Real-time data processing challenges in applications like self-driving cars, drones, and IoT devices
- The impact of 5G networks on the development of real-time applications

Discussion Instructions:

1. Select a real-time system problem related to modern technologies (preferably from a recent paper, industry report, or emerging trend).
2. Describe the problem in detail, explaining its relevance to current computing challenges.
3. Discuss whether this problem has been solved or remains an open issue.
4. If it's still an open problem, propose potential reasons for why it persists and how you think it can be addressed.

My Post:

Hello Class

For this discussion, I chose the integration of AI into real-time decision-making in autonomous systems. In autonomous systems such as self-driving vehicles, drones, and robots, AI not only needs to make a correct decision; it must make the correct decision before a deadline based on real-time, continuously changing data. This kind of time-sensitive computation is known as real-time computing; in real-time computing, correctness depends on both the logical output and the time at which the output is produced (Colorado State University Global, n.d.). For example, a self-driving car detecting a pedestrian too late is almost the same as not detecting the pedestrian at all.

The self-driving car problem can be defined as deadline-aware scheduling for AI in autonomous vehicles. This implies that an autonomous vehicle may need to run several tasks at the same time. The tasks may include processing real-time data from cameras, radar or LiDAR, and various sensors. This real-time data means the system needs to analyze and interpret perception, localization, path planning, braking decisions, cybersecurity monitoring, and communication with infrastructure or other vehicles in real time. These tasks compete heavily for computing resources that are usually limited in such systems, especially for resources such as CPUs and GPUs, and other AI hardware embedded within the vehicle’s overall system.

To solve this complex real-time decision-making problem, autonomous systems such as self-driving cars increasingly depend on deep learning models. Deep learning, which is a type of artificial intelligence that uses layered neural networks to learn patterns from large amounts of data, has improved the perception, object detection, scene understanding, and path planning of autonomous vehicles; however, it also creates new problems related to the models being large, computationally expensive, and difficult to predict in timing behavior (Sahoo & Varadarajan, 2025). A real-time operating system or scheduler for these autonomous systems cannot only ask, "Which task is most accurate?" It must also ask, "Which task must finish first?" and "What happens if the system cannot complete all tasks before their deadlines?"

For the operating system, these questions are mostly related to real-time scheduling issues and real-time system requirements such as determinism, responsiveness, user control, reliability, and fail-soft operation (Colorado State University Global, n.d.). Having a fail-soft operation capability is especially important for autonomous vehicles because a missed deadline could create a safety risk for passengers, pedestrians, or other drivers. In other words, in a situation where the vehicle cannot perform a full AI analysis in time, the vehicle should have backup mechanisms that allow it to continue operating safely. For example, it may need to slow down, use a simpler backup model, or prioritize emergency protocols over less urgent tasks.

However, this real-time scheduling problem related to AI has not been fully solved yet. Yano and Azumi (2025) explain that autonomous driving systems such as Autoware need a guarantee for end-to-end timing. Additionally, self-driving car workloads depend almost exclusively on GPU-bound tasks, and GPU driver implementations and the operating system's limited control over GPU execution time can cause the process to miss deadlines (Zhu et al. 2025).
A possible solution for these real-time scheduling issues, to reduce deadline misses, is to adopt an urgency-aware GPU scheduling approach. However, reducing deadline misses is not the same as eliminating the problem completely.

Additionally, because AI workloads are not always predictable, this real-time scheduling problem persists for several reasons. One of these reasons is that the time required to process sensor data may change based on factors such as traffic, weather, road conditions, and pedestrians crossing the road. Another reason is that AI autonomous systems are heterogeneous by definition. To function properly, they need to access CPUs, GPUs, neural accelerators, sensors, and communication networks all at the same time, making scheduling more complex than just handling CPU scheduling and predicting execution time on one processor. A third reason is that safety systems need strong evidence that deadlines will be met to work properly, not just a theoretical possibility that the deadlines will be met.

To address these issues, a possible solution may be to implement a combination of deadline-aware scheduling techniques, more stable hardware, and more adaptive AI models. This implies that the system should be designed to prioritize urgent and safety tasks. For example, emergency braking, obstacle detection, and collision avoidance should receive higher priority than less urgent tasks.

-Alex

References:

Colorado State University Global. (n.d.). *Lecture 6: Scheduling* [Interactive lecture]. Canvas.

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Zhu, H., Zhang, W., Zhang, X., Tao, Z., Lin, X., Zhang, Y., Ji, J., & Zhang, Y. (2025). *UrgenGo: Urgency-aware transparent GPU kernel launching for autonomous driving*. arXiv. https://doi.org/10.48550/arXiv.2509.12207